

Report Writing Tips

Here are some writing tips to help you write a better report in a journal article style. Please note that the references and models mentioned here are **fictitious**. See the following sentences and paragraphs on the suggested style of writing.

We use the Total Fertility Rate and the Total Life Births data from the Singapore government Department of Statistics website, hereafter abbreviated as TFR and TLB respectively.

A time-series model of the Malaysian Total Fertility Rate from 1955 to 2010 is found Tan and Lim (2024). In their paper, their model is an ARIMA(2,0,1). This is because the Malaysian Total Fertility Rate data over that period is relatively flatter compared to the Singapore Total Fertility Rate.

Johnson et al. (2021) modelled American Total Live Births from 1950 to 2020. Theirs is a seasonal model because they used monthly data.

In Figure 3, we see that the partial autocorrelogram of the first order difference of TFR (over the training part of the data), ∇X_t , has significant spikes all the way to lag 6. Thus our initial model for X_t is ARIMA(6,1,0). The autocorrelogram and partial autocorrelograms of this model's residuals are shown in Figures 7 and 8. They still show marginally significant spikes at lag 16 and 21. This suggests that some moving average terms may need to be added to our initial ARIMA(6,1,0) model. By adjusting the autoregressive and moving average parameters for our initial ARIMA(6,1,0) model, we found that the following viable models ARIMA(6,1,1), ARIMA(6,1,2) and ARIMA(5,1,1) all produce white noise residuals.

A summary of our models for TFR is found in Table 3. In Table 3, the first and second columns list the models and their respective AIC values for the training part (1960 to 2012). The third column lists the sum of square error of the predicted values of each model with respect to the actual observations in the testing part of the data (2013 to 2024¹). We see from Table 3 that the ARIMA(5,1,1) has the smallest AIC value of 201 and it also has the smallest sum of square error² over the testing part. Thus we select the ARIMA(5,1,1) model and the model equation is

$$(\text{Display your equations properly with equation numbers}) \tag{1}$$

where Z_t is Gaussian white noise with distribution $N(0, 0.367)$.

¹Could also be 2025 for some of you, depending on when you downloaded the data.

²You could also use another reasonable metric to measure the fit for the testing part, or even use 2 different metrics, and make comparisons. You might also have conflicting results where one model for the training part gives the lowest AIC and another gives the smallest SSE for the testing part, in which case you can include a discussion on which one could be a better choice

Further Notes

1. Do not copy my paragraphs and sentences on Page 1 word for word, changing only the model specifications and the table/figure numbers.
2. Some of the literature that you have found might not be exactly technical literature, but if you think it will help you and the reader of the report gain further insight into your modelling, you can also include them, summarize their views, and perhaps mention whether these paper will influence the way that you would eventually choose your final model among the viable models with white noise residuals.
3. It is likely that some of you would also take the log values of TFR and TLB as well, so you can have even more tables and comparisons.
4. In the event that you come up with an unexpected model which is still viable (that is, has white noise residuals) but appears to be completely different from the others, you could still include it, note its AIC value, calculate the SSE (or the metric that you use) for the testing part, and then do further comparisons with your other models. You could also try to come up with reasons why this unexpected model works, if you think you have found such a model.